1) $5A - X + 2B = C$

$$A = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & -2 \\ -2 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -2 & 0 \\ -1 & 3 & 0 \\ -2 & 5 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} -1 & -1 & 4 \\ -1 & -2 & 6 \\ 1 & 1 & -5 \end{pmatrix}$$

$$X = (x_{ij}) = (5a_{ij} - 2b_{ij} - c_{ij}) = \begin{pmatrix} 8 & -3 & -4 \\ -11 & 13 & -16 \\ -15 & 9 & 12 \end{pmatrix}$$

2) $P(t) = -9t^3 + t^2 - 4t - 7$, $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 2 & 1 \end{pmatrix}$

$P(A) = -9A^3 + A^2 - 4A - 7I$

$A^2 = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{pmatrix}$

$A^3 = A^2 \cdot A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 2 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 2 \\ 1 & 2 & 1 \\ 2 & 4 & 3 \end{pmatrix}$

$-9A^3 = \begin{pmatrix} -18 & -27 & -18 \\ -9 & -18 & -9 \\ -18 & -36 & -27 \end{pmatrix}$, $-7I = \begin{pmatrix} -7 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 0 & -7 \end{pmatrix}$

$-4A = \begin{pmatrix} 0 & -4 & -4 \\ -4 & 0 & 0 \\ 0 & -8 & -4 \end{pmatrix}$

$P(A) = D = \begin{pmatrix} -24 & -29 & -21 \\ -13 & -24 & -8 \\ -16 & -42 & -34 \end{pmatrix}$

3) $D = -5A - 3B + C$

$$-5A = \begin{pmatrix} -5 & 10 & 10 \\ 10 & -25 & -25 \\ -5 & 10 & 5 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$-3B = \begin{pmatrix} -3 & 6 & 3 \\ 0 & -3 & -3 \\ 3 & 6 & 3 \end{pmatrix}$$

$$D = \begin{pmatrix} -7 & 15 & 12 \\ 10 & -22 & -28 \\ -4 & 16 & 8 \end{pmatrix}$$

4) $\begin{pmatrix} -2 & -1 \\ 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 4 & 2 \\ 0 & 0 \end{pmatrix}$

5) $A_{1\times 4} \cdot B_{m\times n} = C_{1\times 3} \Rightarrow m = 4,\ n = 3$

6) $A_{2\times1} \cdot B_{m\times n} = C_{2\times1} \Rightarrow m=1,\ n=1$

7) $(3\ 2\ 4) \cdot \begin{pmatrix} -1 & 1 \\ -5 & 0 \\ 2 & -4 \end{pmatrix} \cdot \begin{pmatrix} -5 & -1 \\ -4 & -2 \end{pmatrix} =$

$= (3\ 2\ 4) \cdot \begin{pmatrix} 1 & -1 \\ 5 & 0 \\ -2 & 4 \end{pmatrix} \cdot \begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix} =$

$= (5\ 13) \cdot \begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix} = (77\ 31)$

8) $A = \begin{pmatrix} 0 & -3 \\ -1 & -3 \end{pmatrix}$

$A^T = \begin{pmatrix} 0 & -1 \\ -3 & -3 \end{pmatrix}$

9) $\begin{pmatrix} -2 & -1 \\ -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$

10) $\begin{pmatrix} 11 & 1 \\ 0 & 11 \end{pmatrix}^8 = \begin{pmatrix} 11^8 & 8 \cdot 11^7 \\ 0 & 11^8 \end{pmatrix}$

- $\begin{pmatrix} 11 & 1 \\ 0 & 11 \end{pmatrix}^2 = \begin{pmatrix} 11^2 & 22 \\ 0 & 11^2 \end{pmatrix}$

- $\begin{pmatrix} 11 & 1 \\ 0 & 11 \end{pmatrix}^3 = \begin{pmatrix} 11^3 & 11^2 + \overbrace{22 \cdot 11}^{2 \cdot 11^2} = 3 \cdot 11^2 \\ 0 & 11^3 \end{pmatrix} \Rightarrow$

$\Rightarrow \begin{pmatrix} 11 & 1 \\ 0 & 11 \end{pmatrix}^n = \begin{pmatrix} 11^n & n \cdot 11^{n-1} \\ 0 & 11^n \end{pmatrix}$

11) $A = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 1 & 3 \\ 0 & -1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -2 & -2 \\ -2 & 5 & 4 \\ 0 & 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & -2 & -1 \\ 0 & 1 & 0 \\ 1 & -2 & 0 \end{pmatrix}$

$D = 4A + 2B = \begin{pmatrix} 6 & -4 & -8 \\ -12 & 14 & 20 \\ 0 & -4 & 2 \end{pmatrix}$